Hospital Billing and Patient Classification – Complete Python Solution

Problem Statement:

Develop a Python program to process hospital bills by calculating consultation charges, diagnostic charges, and treatment charges. Apply conditional concessions, validate billing constraints, calculate the final payable amount, and classify patients according to a supplied payment policy.

The solution uses only your assigned Python syllabus concepts.

1. Computational Thinking

Input:

- Patient name
- Patient ID
- Consultation charges
- Diagnostic charges
- Treatment charges
- Payment amount

Processing:

1. Read patient details.
2. Read consultation, diagnostic, and treatment charges.
3. Validate all billing amounts.
4. Calculate the total bill.
5. Apply conditional concessions.
6. Calculate the final payable amount.
7. Calculate the outstanding balance.
8. Classify the patient according to the payment policy.
9. Generate the hospital bill statement.

Output:

- Patient details
- Individual charges
- Total bill
- Concession amount

- Final bill
- Amount paid
- Outstanding balance
- Patient payment classification

2. Hospital Billing Policy

For this example, assume the following payment rules:

Conditional concession policy

Total bill amount	Concession
Below ₹5,000	0%
₹5,000–₹9,999	5%
₹10,000–₹19,999	10%
₹20,000 and above	15%

Patient payment classification

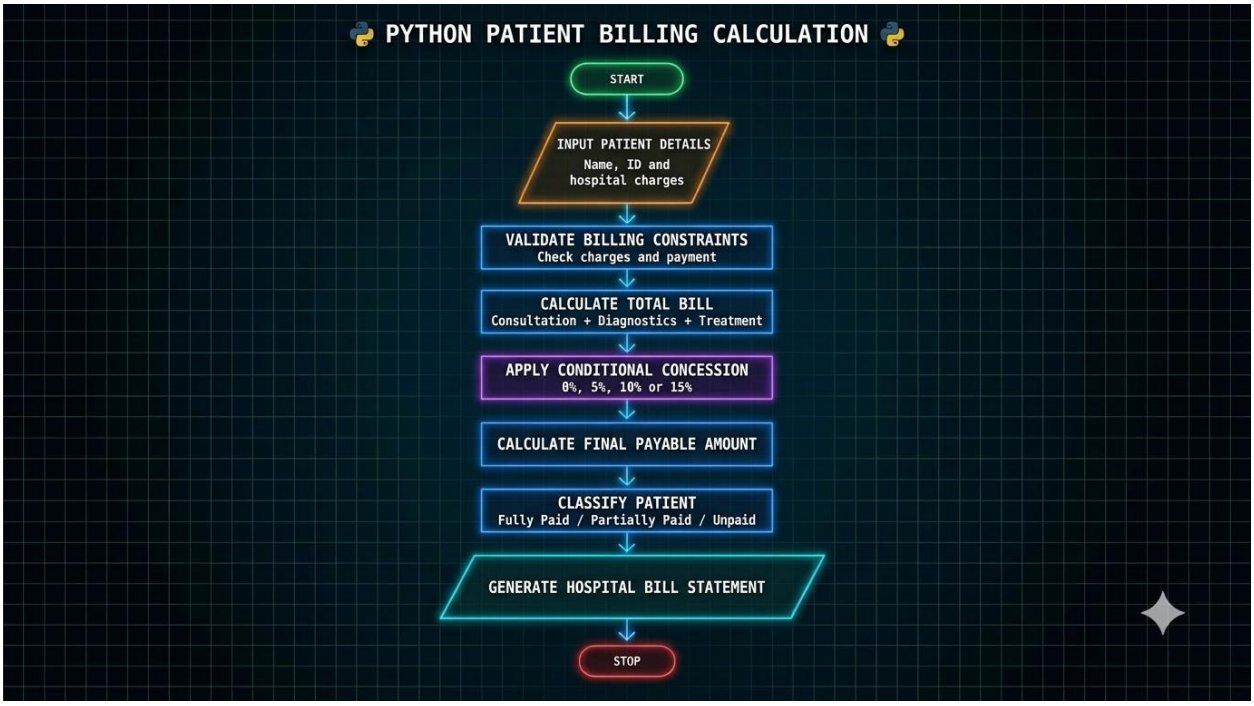
Payment condition	Classification
Full payment completed	Fully Paid
Partial payment made	Partially Paid
No payment made	Unpaid
Payment exceeds final bill	Invalid Payment

3. Algorithm

1. Start.
2. Read patient name and patient ID.
3. Read consultation charges.
4. Read diagnostic charges.
5. Read treatment charges.
6. Read payment amount.
7. Validate that all charges are non-negative.
8. Calculate total bill.

9. Apply the appropriate concession percentage.
10. Calculate concession amount.
11. Calculate final payable amount.
12. Validate payment amount.
13. Calculate outstanding balance.
14. Classify the patient.
15. Display the formatted hospital bill.
16. Stop.

4. Flowchart



5. Pseudocode

START

INPUT patient_name

INPUT patient_id

INPUT consultation_charge

```
INPUT diagnostic_charge
INPUT treatment_charge

INPUT payment_amount

IF any charge < 0 OR payment_amount < 0:
   DISPLAY "Invalid Billing Details"

ELSE:

   total_bill = consultation_charge
              + diagnostic_charge
              + treatment_charge

   IF total_bill < 5000:
      concession_rate = 0

   ELSE IF total_bill < 10000:
      concession_rate = 0.05

   ELSE IF total_bill < 20000:
      concession_rate = 0.10

   ELSE:
      concession_rate = 0.15

   concession = total_bill * concession_rate
```

final_bill = total_bill - concession

IF payment_amount > final_bill:

DISPLAY "Invalid Payment"

ELSE:

balance = final_bill - payment_amount

IF payment_amount == final_bill:

classification = "FULLY PAID"

ELSE IF payment_amount > 0:

classification = "PARTIALLY PAID"

ELSE:

classification = "UNPAID"

DISPLAY hospital bill statement

STOP

6. Complete Python Program

```
# Hospital Billing and Patient Classification

# Input patient details
patient_name = input("Enter patient name: ")
patient_id = input("Enter patient ID: ")
```

```
# Input hospital charges
consultation = float(input("Enter consultation charge: "))
diagnostic = float(input("Enter diagnostic charge: "))
treatment = float(input("Enter treatment charge: "))

# Input payment details
payment = float(input("Enter payment amount: "))

# Validate billing constraints
if (consultation < 0 or diagnostic < 0 or
    treatment < 0 or payment < 0):

  print("\nInvalid Billing Details!")
  print("Charges and payment cannot be negative.")

else:

  # Calculate total bill
  total_bill = consultation + diagnostic + treatment

  # Apply conditional concessions
  if total_bill < 5000:
    concession_rate = 0

  elif total_bill < 10000:
    concession_rate = 0.05
```

```
elif total_bill < 20000:
    concession_rate = 0.10

else:
    concession_rate = 0.15

# Calculate concession
concession = total_bill * concession_rate

# Calculate final bill
final_bill = total_bill - concession

# Validate payment amount
if payment > final_bill:

    print("\nInvalid Payment!")
    print("Payment exceeds the final bill amount.")

else:

    # Calculate outstanding balance
    balance = final_bill - payment

    # Patient classification
    if payment == final_bill:
```

```
    classification = "FULLY PAID"

elif payment > 0:
    classification = "PARTIALLY PAID"

else:
    classification = "UNPAID"

# Store billing details in a list
billing_details = [
    consultation,
    diagnostic,
    treatment,
    total_bill,
    concession,
    final_bill,
    payment,
    balance
]

# Display hospital bill statement
print("\n====================================")
print("       HOSPITAL BILL STATEMENT")
print("====================================")

print("Patient Name      :", patient_name)
```

```
print("Patient ID        :", patient_id)

print("-------------------------------------")

print("Consultation      : Rs.",
    round(billing_details[0], 2))

print("Diagnostic        : Rs.",
    round(billing_details[1], 2))

print("Treatment         : Rs.",
    round(billing_details[2], 2))

print("-------------------------------------")

print("Total Bill        : Rs.",
    round(billing_details[3], 2))

print("Concession        : Rs.",
    round(billing_details[4], 2))

print("Final Bill        : Rs.",
    round(billing_details[5], 2))

print("Amount Paid       : Rs.",
    round(billing_details[6], 2))
```

```
    print("Outstanding Balance: Rs.",
        round(billing_details[7], 2))

    print("-------------------------------------")

    print("Patient Classification:", classification)

    print("=====================================")
    print("      BILLING COMPLETED")
```

7. Sample Input

Enter patient name: Rahul

Enter patient ID: H101

Enter consultation charge: 2000

Enter diagnostic charge: 4000

Enter treatment charge: 6000

Enter payment amount: 9000

8. Sample Output

```
====================================
      HOSPITAL BILL STATEMENT
====================================

Patient Name      : Rahul
Patient ID        : H101
```

Consultation : Rs. 2000.0

Diagnostic : Rs. 4000.0

Treatment : Rs. 6000.0

Total Bill : Rs. 12000.0

Concession : Rs. 1200.0

Final Bill : Rs. 10800.0

Amount Paid : Rs. 9000.0

Outstanding Balance: Rs. 1800.0

Patient Classification: PARTIALLY PAID

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BILLING COMPLETED

9. Concession Calculation

For the above example:

- Consultation = ₹2,000
- Diagnostic = ₹4,000
- Treatment = ₹6,000

Step 1: Calculate total bill

\[2000+4000+6000=12000 \]

Step 2: Apply concession

Since the total bill is ₹12,000, the concession rate is 10%.

$$ 12000\times\frac{10}{100}=1200 $$

Step 3: Calculate final bill

$$ 12000-1200=10800 $$

Step 4: Calculate outstanding balance

$$ 10800-9000=1800 $$

Final classification: PARTIALLY PAID.

10. Additional Test Cases

Total bill	Payment	Classification
₹4,000	₹4,000	Fully Paid
₹8,000	₹5,000	Partially Paid
₹12,000	₹0	Unpaid
₹25,000	₹25,000	Fully Paid
₹10,000	₹12,000	Invalid Payment

11. Assigned Python Syllabus Concepts Used

Syllabus concept	Application
Computational thinking	Breaking billing into steps
Algorithm	Step-by-step calculation
Flowchart	Program execution flow
Pseudocode	Logical representation
Variables	Charges, payment, concession
Data types	String and float
Operators	+, -, *, <, >, ==
Conditional statements	if, elif, else
Strings	Patient name and classification
Lists	Store billing details

Built-in functions	input(), print(), round()

Complete syllabus-based solution

Important: The concession percentages and payment classifications are illustrative policies for this programming exercise, not actual hospital regulations.